

Clinically Interpretable Validation of Mechanistic Cardiovascular Biomarkers Across Demographic and Physiological Subgroups

Arjun Krishnamurthy^{1,*†}, Anishsairam Murari^{1,*}, Anshul Palreddy^{1,*}, Aayush Pal^{1,*}

¹Texas Academy of Mathematics and Science, University of North Texas, Denton, TX

[†]Correspondence: arjun.krish01x@gmail.com

Abstract

Background: Interpretable cardiovascular biomarkers such as blood pressure, cholesterol, glucose, and body mass index are often treated as clinically trustworthy because they correspond to familiar physiology. However, physiologic interpretability alone does not establish that a prediction model remains discriminative, calibrated, equitable, or clinically useful across patient subgroups. **Objective:** We evaluated whether an interpretable cardiovascular biomarker panel maintained reliable performance across demographic and physiologic strata in a large real-world cohort. **Methods:** A public cardiovascular disease dataset was filtered for physiologic plausibility, yielding 68,599 observations with 33,934 cardiovascular-disease-positive cases. Logistic regression and random forest models were trained using systolic blood pressure, diastolic blood pressure, pulse pressure, mean arterial pressure, cholesterol, fasting glucose, body mass index, and lifestyle covariates. Model performance was assessed using held-out AUC, Expected Calibration Error, subgroup disparity metrics, Population Stability Index, Kullback-Leibler divergence, permutation importance, and Decision Curve Analysis. **Results:** The random forest achieved an overall AUC of 0.784 (95% CI: 0.776-0.790). Subgroup AUC ranged from 0.592 (95% CI: 0.544-0.633) in Baseline BP category: Elevated BP to 0.820 (95% CI: 0.800-0.841) in Age group: Age ≥ 45 , with a maximum baseline BP-category AUC gap of 0.127 (95% CI: 0.091-0.176). Age-related disparities exceeded sex-related disparities, with demographic parity difference of 0.297 (95% CI: 0.277-0.316) and equalized odds difference of 0.217 (95% CI: 0.197-0.236). **Conclusion:** Interpretable cardiovascular biomarkers can support aggregate prediction while still exhibiting subgroup-specific degradation in discrimination, calibration, stability, and decision utility. These findings support routine subgroup validation before clinical translation of cardiovascular risk models.

1 Introduction

Cardiovascular disease remains a leading source of morbidity, mortality, and health-system burden, making accurate risk stratification a central goal of biomedical engineering, preventive cardiology, and clinical decision support [8]. Machine learning models are increasingly used to combine demographic, physiologic, laboratory, and lifestyle variables into individualized risk estimates [4]. In cardiovascular applications, many commonly used predictors are clinically interpretable: systolic blood pressure reflects vascular load, pulse pressure approximates arterial stiffness, cholesterol and fasting glucose represent cardiometabolic risk, and body mass index captures obesity-related burden. This physiologic familiarity can make such models appear more trustworthy than black-box systems that rely on opaque inputs.

However, clinical interpretability is not equivalent to clinical reliability. A model can achieve acceptable aggregate discrimination while failing in a subgroup, producing miscalibrated probabilities, amplifying demographic performance gaps, or offering little net benefit at the decision thresholds used for screening and triage [5, 12]. These issues are especially important for cardiovascular prediction because age, sex, blood pressure distribution, medication exposure, and cardiometabolic risk factors can shift the relationship between biomarkers and outcome labels [9]. Therefore, validation must move beyond overall AUC and evaluate whether interpretable biomarkers remain stable across clinically meaningful strata.

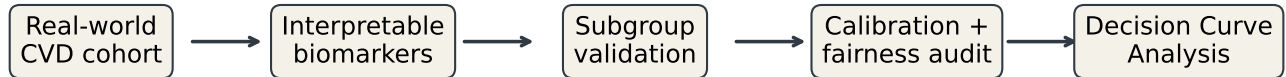


Figure 1: **Cardiovascular biomarker validation workflow.** The pipeline begins with a real-world cardiovascular cohort, constructs interpretable physiologic and lifestyle biomarkers, evaluates subgroup discrimination and calibration, quantifies fairness and distribution shift, and estimates clinical utility using Decision Curve Analysis.

Prior work on clinical machine learning has emphasized the need for transparent reporting, external validation, calibration assessment, and evaluation of bias before deployment [6, 7, 11]. Similarly, Decision Curve Analysis provides a clinically interpretable method for determining whether predicted risks improve decisions compared with default strategies such as treating all or treating none [3]. Despite these advances, many cardiovascular machine learning studies still emphasize average model performance and provide limited evidence about subgroup stability, calibration, distribution shift, or decision utility.

In this study, we evaluate a subgroup validation framework for interpretable cardiovascular biomarkers in a large real-world tabular cohort. The objective is not to introduce a new model architecture. Instead, we test whether a biomarker panel composed of blood pressure, cholesterol, glucose, body mass index, and lifestyle covariates remains discriminative, calibrated, fair, stable, and clinically useful across age, sex, and baseline blood pressure subgroups. We hypothesized that aggregate model performance would obscure subgroup-specific weaknesses, particularly across age and baseline blood pressure categories. Figure 1 summarizes the proposed validation pipeline, which links real-world cardiovascular data, interpretable biomarker extraction, subgroup validation, calibration and fairness auditing, and clinical utility analysis.

2 Methods

2.1 Dataset and Study Design

We conducted a retrospective validation study using a publicly available cardiovascular disease dataset containing 70,000 records. Each record included age, sex, height, weight, systolic blood pressure, diastolic blood pressure, cholesterol category, fasting glucose category, smoking status, alcohol use, physical activity, and binary cardiovascular disease label [1]. The dataset was analyzed as a de-identified public research dataset; no direct patient identifiers were available.

Records were excluded if they contained physiologically implausible values according to predefined criteria: age outside 18–100 years, systolic blood pressure outside 80–240 mmHg, diastolic blood pressure outside 40–140 mmHg, height outside 120–220 cm, weight outside 30–250 kg, body mass index outside 12–70 kg/m², or systolic blood pressure less than or equal to diastolic blood pressure. These thresholds were specified before model evaluation to reduce measurement artifacts while preserving clinically plausible variation. After filtering, the final analytic cohort contained 68,599 observations, including 33,934 cardiovascular-disease-positive cases. No missing values were present after preprocessing.

2.2 Biomarker Construction

The evaluated biomarker set included directly measured variables and derived physiologic variables. Body mass index was calculated as weight in kilograms divided by squared height in meters. Pulse pressure was calculated as systolic blood pressure minus diastolic blood pressure. Mean arterial pressure was estimated as diastolic blood pressure plus one-third of pulse pressure. Cholesterol and fasting glucose were encoded as ordinal categorical variables according to the dataset coding scheme. Smoking, alcohol use, and physical activity were treated as binary lifestyle covariates.

The final model feature set included systolic blood pressure, diastolic blood pressure, pulse pressure, mean arterial pressure, cholesterol category, fasting glucose category, body mass index, smoking status, alcohol use, and physical activity. Maximum heart rate, resting ECG status, and exercise-induced angina were not available in this dataset and were not imputed.

2.3 Subgroup Definitions

Subgroup analyses were performed across age group, sex, and baseline blood pressure category. Age was converted from days to years and categorized as ≤ 45 , 46–55, and 56–65 years; no participants older than 65 years remained in the source cohort after preprocessing. Sex was analyzed using the dataset-provided binary coding and reported as male and female according to the dataset documentation. Baseline blood pressure categories were derived from systolic and diastolic blood pressure using guideline-aligned thresholds: normal, elevated, stage 1 hypertension, and stage 2 hypertension [9]. Subgroup sizes, outcome prevalence, discrimination, and calibration metrics are reported in Table 1.

2.4 Model Development

The cohort was split into training and testing sets using a stratified 75/25 split to preserve outcome prevalence. Logistic regression and random forest models were trained using the same interpretable biomarker set. Logistic regression served as a linear baseline, while random forest modeling captured nonlinear associations and interactions among physiologic predictors without introducing non-interpretable input modalities.

The random forest used 400 trees, maximum depth unrestricted, minimum samples per leaf 20, minimum samples per split 2, and class weighting set to balanced subsampling. The random seed was fixed at 42. Continuous variables were standardized for logistic regression and used in native scale for random forest. Ordinal and binary covariates were encoded using the dataset-provided numeric coding. No hyperparameter tuning was performed on the test set; all reported metrics were computed once on the held-out test set.

2.5 Discrimination and Calibration

Model discrimination was quantified using area under the receiver operating characteristic curve. AUC was computed for the full test set and for each predefined subgroup. Subgroup AUC was reported only when both outcome classes were present. Confidence intervals were computed using 300 stratified bootstrap replicates of the held-out test set [10]. Calibration was evaluated using quantile-binned calibration curves and Expected Calibration Error. For Expected Calibration Error, predictions were divided into 10 bins, and the weighted average absolute difference between predicted risk and observed event rate was computed across bins [11].

2.6 Fairness and Subgroup Disparity Metrics

Fairness metrics were evaluated descriptively across age and sex strata using definitions aligned with Fairlearn terminology [2]. Demographic parity difference was defined as the maximum absolute difference

in positive prediction rates across subgroup levels. Equalized odds difference was defined as the maximum subgroup difference in true positive rates and false positive rates [12]. Because these metrics were computed from observational data, they were interpreted as subgroup disparity checks rather than causal estimates of algorithmic bias. The classification threshold for fairness metrics was 0.50.

2.7 Feature Stability and Distribution Shift

Feature stability was evaluated using Population Stability Index and Kullback-Leibler divergence. For each biomarker, subgroup distributions were compared with the full cohort distribution using decile-based bins. Higher Population Stability Index or Kullback-Leibler divergence indicated greater distributional shift across age or baseline blood pressure strata.

2.8 Clinical Utility

Decision Curve Analysis was used to estimate net benefit across threshold probabilities from 0.05 to 0.80 [3]. Net benefit was evaluated for the overall model, selected baseline blood pressure subgroups, and default treat-all and treat-none strategies. Clinically relevant threshold probabilities of 0.10, 0.20, 0.40, and 0.60 were stored in the analysis outputs.

2.9 Reproducibility, Ethics, and Availability

The analysis used Python 3.14.4, NumPy 2.4.6, pandas 3.0.3, scikit-learn 1.8.0, and matplotlib 3.10.9. Random seed, package versions, model hyperparameters, missingness counts, and filtering counts are stored with the generated tables. The dataset is publicly available and de-identified; no institutional review board approval was sought for this secondary analysis of public data. Code and generated analysis artifacts are included with the submitted project package.

3 Results

3.1 Cohort and Overall Model Performance

After physiologic plausibility filtering, the analytic cohort contained 68,599 observations with 33,934 cardiovascular-disease-positive cases (49.5%). The held-out test set contained 17,150 observations with 8,484 positive cases. Table 1 reports subgroup sample sizes, outcome prevalence, discrimination, and calibration metrics.

The random forest achieved an overall test AUC of 0.784 (95% CI: 0.776-0.790), compared with logistic regression AUC of 0.781 (95% CI: 0.774-0.788). Because the random forest showed slightly stronger held-out discrimination while using the same interpretable biomarker set, subsequent subgroup validation focused on random forest predictions. The full validation workflow is shown in Fig. 1.

3.2 Subgroup Discrimination

Subgroup AUC varied across demographic and physiologic strata (Fig. 2A). The highest observed AUC was 0.820 (95% CI: 0.800-0.841) in Age group: Age ≥ 45 , whereas the lowest observed AUC was 0.592 (95% CI: 0.544-0.633) in Baseline BP category: Elevated BP. The largest subgroup AUC gap was 0.127 (95% CI: 0.091-0.176) across baseline blood pressure categories (Table 2). These findings indicate that overall model discrimination obscured subgroup heterogeneity relevant to pre-deployment validation.

3.3 Calibration and Fairness

Calibration curves demonstrated subgroup-dependent differences between predicted risk and observed cardiovascular disease frequency (Fig. 2B). The highest subgroup Expected Calibration Error was 0.110

Table 1: Cohort and subgroup characteristics with held-out model performance.

Group	N	CVD-positive	AUC (95% CI)	ECE (95% CI)
Overall cohort	17,150	8,484 (49.5%)	0.784 (0.776-0.790)	0.011 (0.008-0.017)
Age group: Age 46-55	6,956	3,125 (44.9%)	0.793 (0.783-0.803)	0.032 (0.026-0.042)
Age group: Age 56-65	7,743	4,658 (60.2%)	0.734 (0.723-0.745)	0.054 (0.049-0.064)
Age group: Age $\geq$ 45	2,451	701 (28.6%)	0.820 (0.800-0.841)	0.110 (0.102-0.121)
Sex: Female	11,117	5,484 (49.3%)	0.786 (0.778-0.794)	0.010 (0.009-0.019)
Sex: Male	6,033	3,000 (49.7%)	0.780 (0.767-0.791)	0.015 (0.012-0.028)
Baseline BP category: Elevated BP	787	257 (32.7%)	0.592 (0.544-0.633)	0.036 (0.019-0.071)
Baseline BP category: Normal BP	2,361	507 (21.5%)	0.624 (0.596-0.647)	0.016 (0.011-0.034)
Baseline BP category: Stage 1 BP	9,976	4,487 (45.0%)	0.719 (0.710-0.730)	0.013 (0.009-0.022)
Baseline BP category: Stage 2 BP	4,026	3,233 (80.3%)	0.649 (0.623-0.670)	0.020 (0.014-0.030)

(95% CI: 0.102-0.121) in Age group: Age $\geq$ 45. Sex-related disparity metrics were small, with demographic parity difference of 0.000 (95% CI: 0.000-0.018) and equalized odds difference of 0.003 (95% CI: 0.002-0.026). In contrast, age-related disparities were larger, with demographic parity difference of 0.297 (95% CI: 0.277-0.316) and equalized odds difference of 0.217 (95% CI: 0.197-0.236) (Table 2). These fairness metrics were interpreted descriptively because subgroup membership may reflect underlying disease prevalence, measurement patterns, and cohort structure.

3.4 Feature Importance and Distributional Stability

Permutation importance identified Systolic BP, Cholesterol, Mean arterial pressure as the strongest predictors. However, feature stability analysis showed that blood-pressure-derived variables were also among the most distributionally unstable biomarkers by maximum Population Stability Index (Fig. 2C). Diastolic BP, Systolic BP, Mean arterial pressure showed the largest subgroup distribution shifts. This combination of high predictive importance and high instability suggests that blood-pressure-derived predictors should be monitored carefully during subgroup-specific translation.

3.5 Clinical Utility

Decision Curve Analysis showed that net benefit varied by baseline blood pressure subgroup across threshold probabilities from 0.05 to 0.80 (Fig. 2D). At intermediate decision thresholds, net benefit differed across normal, elevated, stage 1, and stage 2 blood pressure strata. The stage 2 subgroup showed higher apparent net benefit across much of the threshold range, likely reflecting higher baseline risk and outcome prevalence, whereas the elevated blood pressure subgroup showed lower net benefit. These results indicate that a single aggregate decision threshold may not provide equal clinical utility across physiologic subgroups.

4 Discussion

This study demonstrates that physiologically interpretable cardiovascular biomarkers are not automatically reliable across demographic and physiologic subgroups. Although the random forest achieved moderate aggregate discrimination, subgroup validation revealed heterogeneity in AUC, calibration, fairness metrics, distribution shift, and decision-curve net benefit. The main engineering contribution is a validation workflow that evaluates not only whether interpretable biomarkers predict cardiovascular disease, but also whether their predictive behavior remains stable across populations likely to be encountered in clinical use.

Table 2: Subgroup disparity and stability summary.

Metric	Grouping variable	Estimate	95% CI	Interpretation
AUC gap	Baseline BP category	0.127	0.091-0.176	Largest discrimination disparity
Demographic parity difference	Age group	0.297	0.277-0.316	Larger age-related output disparity
Equalized odds difference	Age group	0.217	0.197-0.236	Larger age-related error-rate disparity
Demographic parity difference	Sex	0.000	0.000-0.018	Minimal sex-related output disparity
Equalized odds difference	Sex	0.003	0.002-0.026	Minimal sex-related error-rate disparity
Maximum PSI	Diastolic BP	11.617	NA	High distribution shift
Maximum PSI	Systolic BP	11.461	NA	High distribution shift
Maximum PSI	Mean arterial pressure	10.785	NA	High distribution shift

The largest discrimination gap occurred across baseline blood pressure categories. This finding is important because blood pressure is both a central cardiovascular risk marker and a variable that can differ substantially by age, measurement context, treatment status, and underlying vascular physiology. The elevated blood pressure subgroup had the lowest observed AUC, suggesting that risk separation may be more difficult in participants whose blood pressure values are abnormal but not yet in higher hypertension stages. In contrast, stage 2 blood pressure strata showed higher net benefit in Decision Curve Analysis, likely reflecting higher baseline risk and clearer risk separation. These results support subgroup-aware interpretation of cardiovascular risk predictions rather than reliance on a single aggregate AUC.

Calibration and fairness analyses further showed that model performance differed more across age than sex. The age-group demographic parity and equalized odds gaps should not be interpreted as proof of causal algorithmic bias, because age is strongly associated with cardiovascular disease prevalence and biomarker distributions. However, these metrics identify where model outputs may behave differently across clinically meaningful groups. In practice, such findings would motivate subgroup-specific calibration, threshold sensitivity analysis, and prospective monitoring before deployment.

The feature stability analysis highlights an additional translational issue. Systolic blood pressure, cholesterol, and mean arterial pressure were among the most predictive features, but blood-pressure-derived variables also showed high distributional instability. This pattern is a common deployment challenge: the features that drive prediction may also be the features most sensitive to subgroup composition, measurement protocols, or case-mix shift. A clinically useful model should therefore be evaluated not only by feature importance but also by whether important predictors remain stable across the populations in which the model will be used.

This work should be interpreted as a validation audit rather than a finished clinical decision system. The dataset supports large-scale analysis of interpretable cardiovascular biomarkers, but it does not include medication use, race or ethnicity, longitudinal follow-up, ECG findings, stress-test variables, maximum heart rate, or externally adjudicated outcomes. Therefore, the results should not be used to guide patient care without external validation in clinically curated cohorts. Future work should apply the same validation framework to EHR-derived and stress-test cohorts, evaluate prospective calibration drift, and test whether subgroup-specific recalibration improves clinical utility.

Overall, the findings support a central BMES-relevant message: biomedical interpretability must be paired with engineering validation. A biomarker panel can be physiologically meaningful and still fail to provide stable, calibrated, equitable, or useful predictions across subgroups. Subgroup validation should therefore be treated as a required step in the translational pathway for cardiovascular machine learning

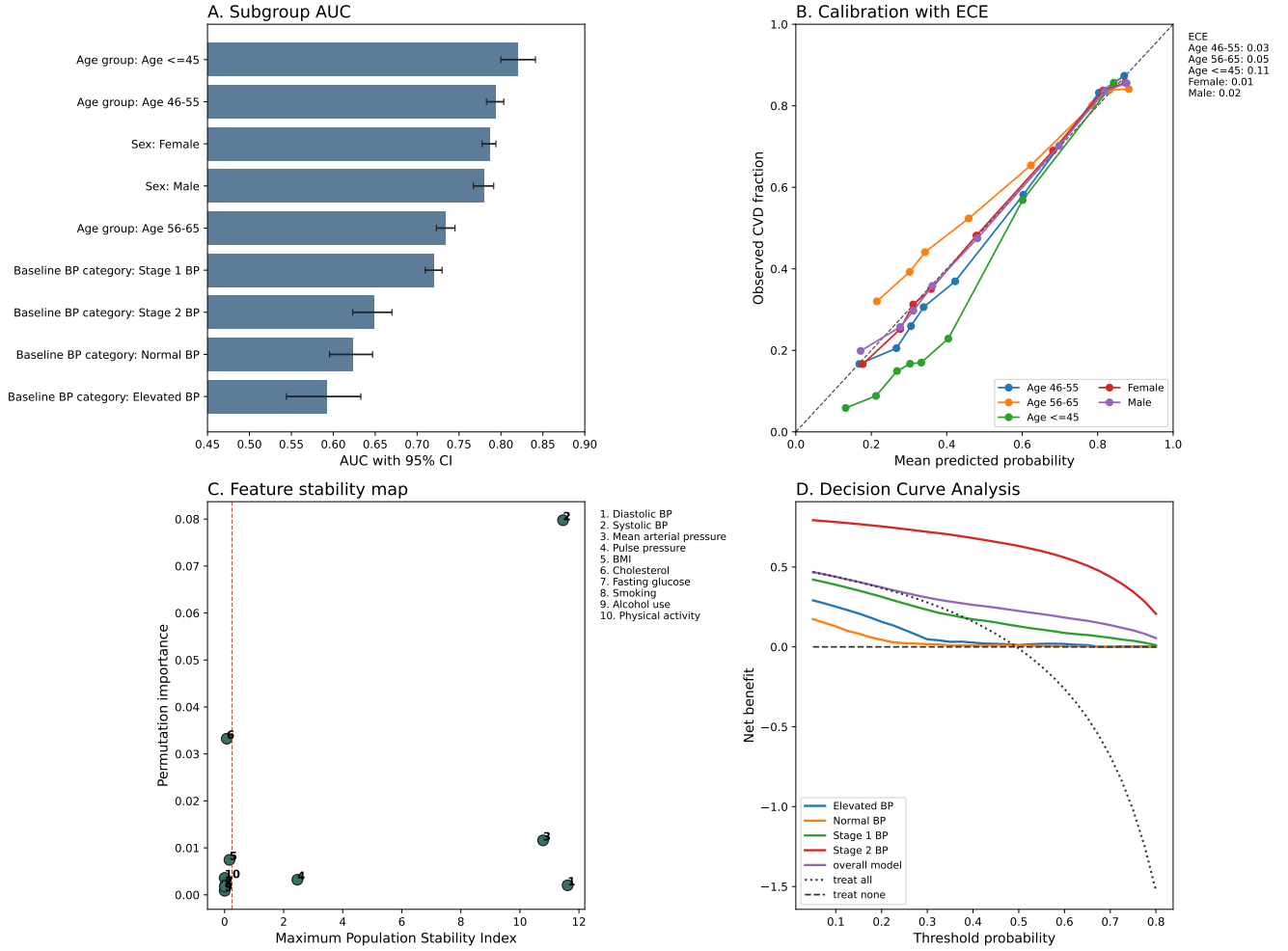


Figure 2: **Subgroup validation of interpretable cardiovascular biomarkers.** (A) Held-out AUC across age, sex, and baseline blood pressure subgroups. (B) Calibration curves with Expected Calibration Error annotations. (C) Feature stability map showing maximum Population Stability Index versus permutation importance. (D) Decision Curve Analysis showing net benefit across threshold probabilities for the full cohort and baseline blood pressure subgroups.

models.

5 Limitations

This study has several limitations. First, the dataset is large but public and tabular, with limited information about data provenance, measurement protocol, medication use, comorbidities, race or ethnicity, socioeconomic variables, and longitudinal follow-up. Second, the cardiovascular disease label should be interpreted as a dataset outcome label rather than a prospectively adjudicated clinical endpoint. Third, the dataset does not include maximum heart rate, resting ECG status, exercise-induced angina, medication exposure, imaging variables, or stress-test findings; therefore, the present analysis validates a limited interpretable biomarker subset rather than a complete cardiovascular diagnostic panel. Fourth, fairness metrics were computed only across available demographic variables and should be interpreted as descriptive subgroup disparity measures, not causal estimates of discrimination or bias. Fifth, all results were obtained from a single dataset without external validation, so model performance may not generalize to other hospitals, countries, measurement protocols, or patient populations. Finally, Decision Curve

Analysis depends on outcome prevalence and threshold selection; prospective clinical studies would be required to determine whether model-guided decisions improve patient outcomes.

6 Conclusion

In a 68,599-record cardiovascular cohort, interpretable biomarkers supported moderate aggregate prediction but showed subgroup variation in discrimination, calibration, fairness metrics, distributional stability, and clinical net benefit. The results demonstrate that physiologic interpretability alone is insufficient for deployment readiness. Cardiovascular risk models should undergo subgroup validation, calibration assessment, feature stability analysis, and decision-curve evaluation before clinical translation. Future work should extend this framework to externally validated EHR, ECG, and stress-test cohorts with richer clinical variables and prospective outcome ascertainment.

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